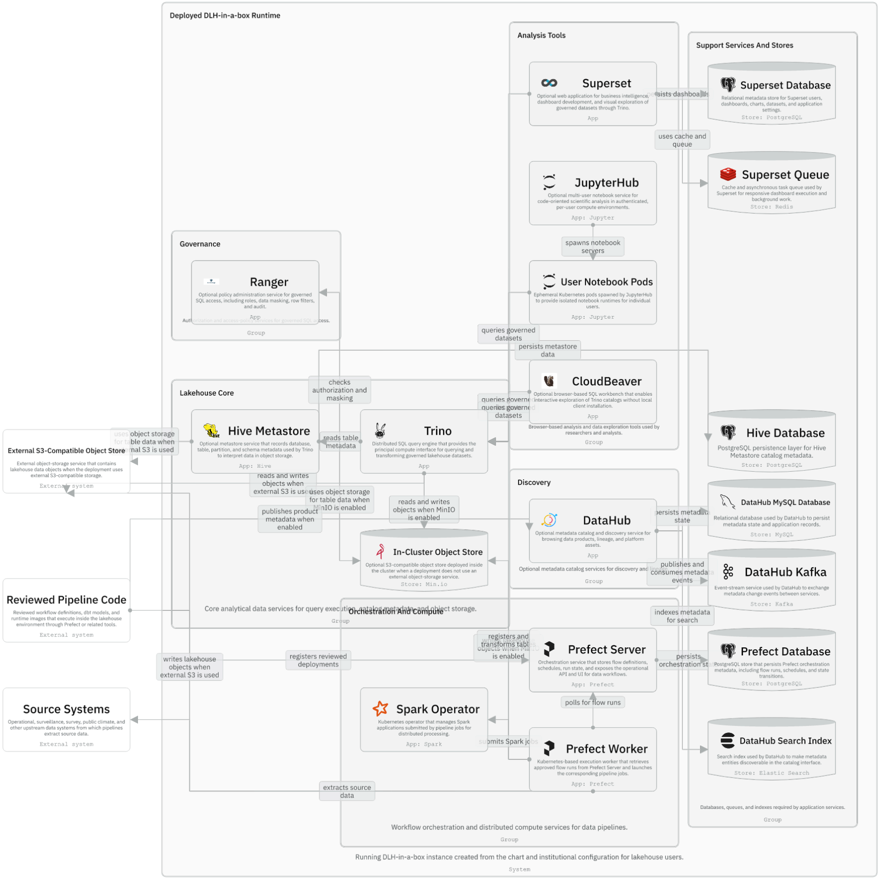
Session 07 / Data plane and surrounding tools

Query, Storage, Metadata, And App Runtime

One hour handover deck

40-45 min narrative / 10-15 min checkpoint / 5 min Q&A



global.dataCatalogs is the shared input that becomes Hive resources, Trino catalogs, governance validation, and Ranger policy inputs.

Session path

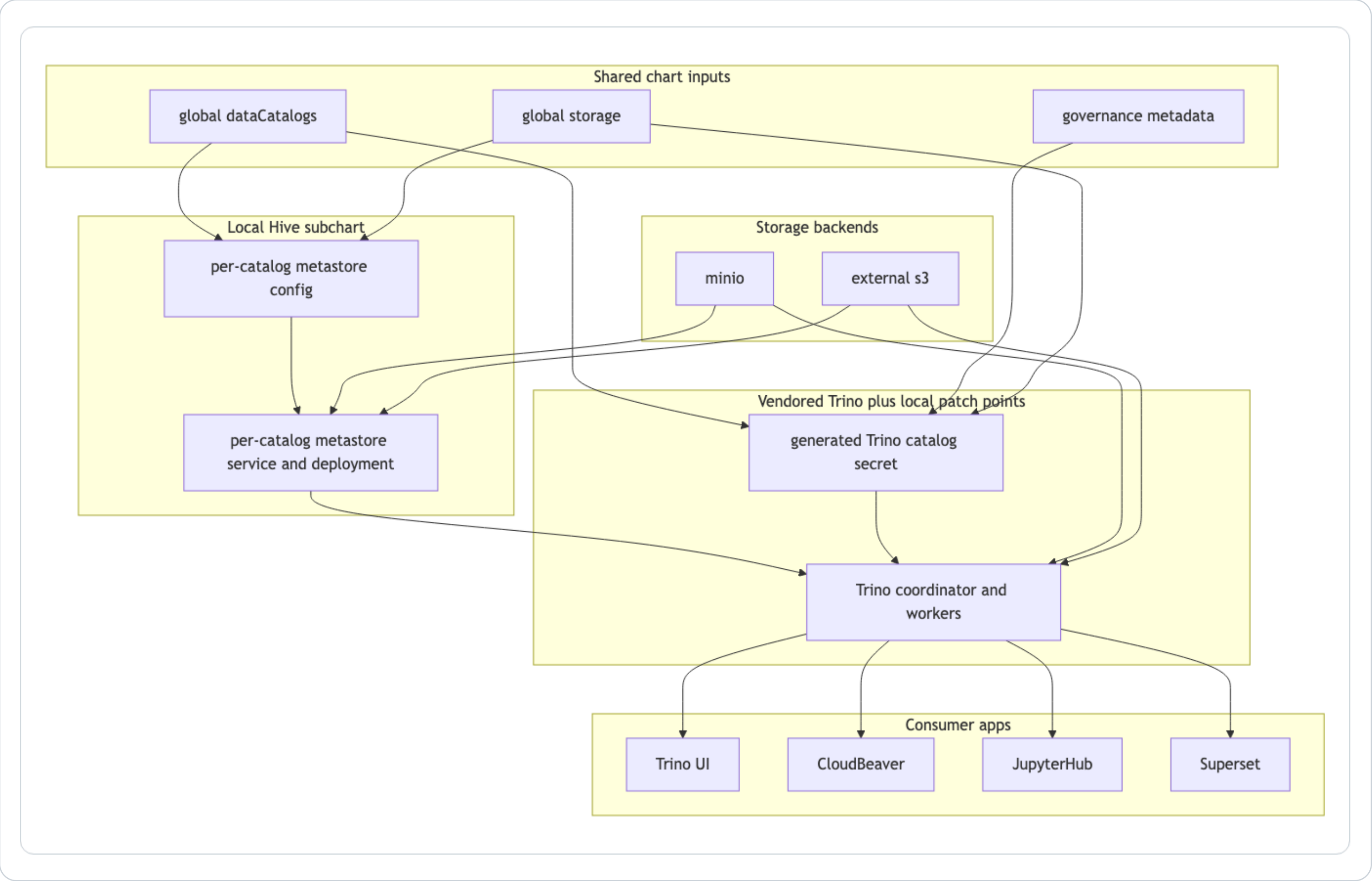
1. Data path
2. Hive subchart
3. Trino patch points
4. Storage modes
5. Consumer apps

Session rhythm

40-45 min narrative
10-15 min checkpoint
5 min Q&A

Presenter notes carry the script and demo prompts.

Data path from values to consumers



- Catalogs drive Hive configuration and Trino catalog secrets.
- Storage settings wire Hive and Trino to MinIO or external S3.
- Governance metadata also feeds validation and access configuration.
- Tools consume data through Trino rather than each inventing a separate path.

Source: manual Data Path Diagram.

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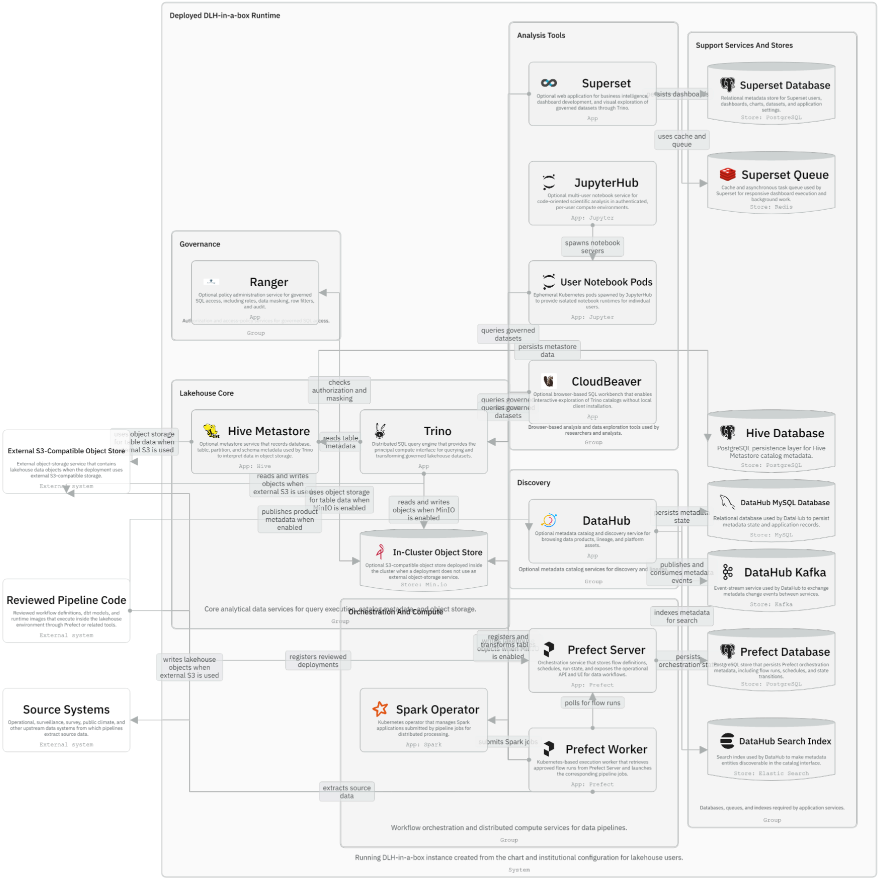
Runtime data, analysis, and orchestration



- Prefect workers run pipelines and can submit Spark jobs.
- Trino queries data and registers/transforms tables.
- CloudBeaver, JupyterHub, and Superset query through Trino.
- DataHub is optional discovery, not the query engine.

Source: IcePanel 11-runtime-data-analysis-orchestration.

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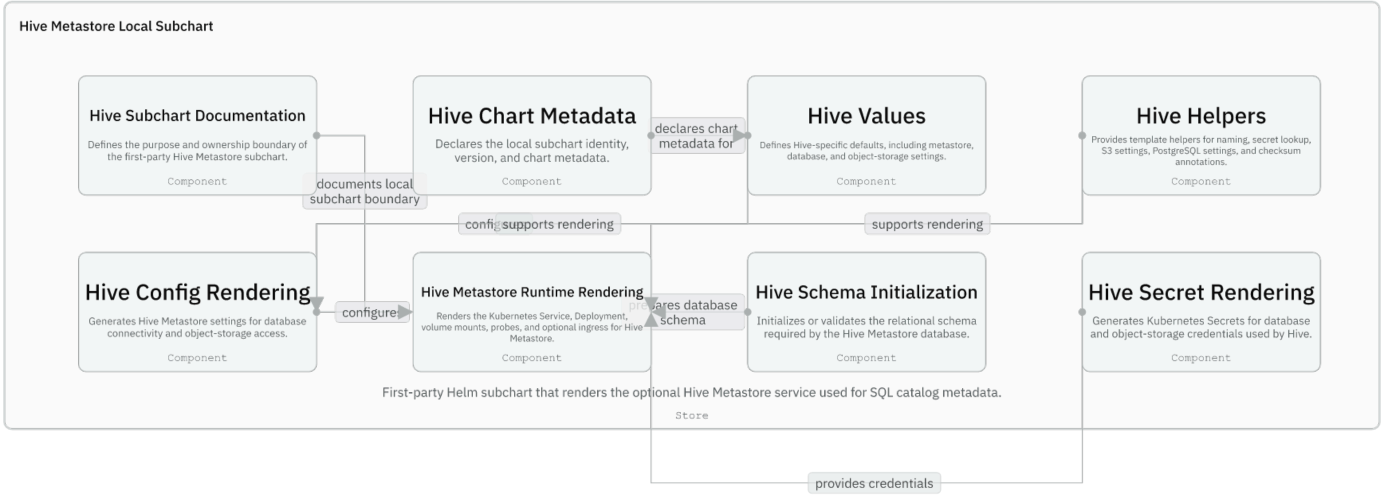
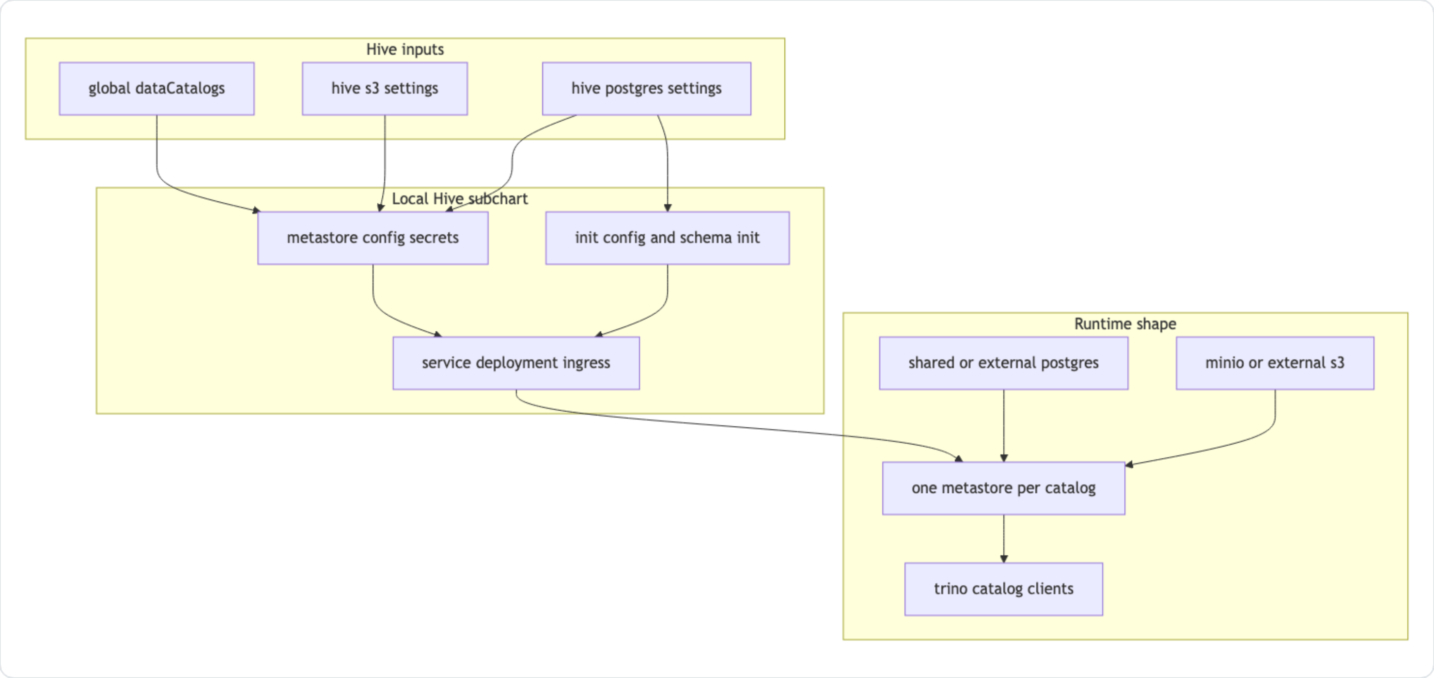
Hive local subchart



- The Hive subchart is fully repo-owned.
- It can render one metastore service and deployment per catalog.
- It owns metastore config, database credentials, object storage credentials, and schema initialization.

Source: IcePanel 07-hive-metastore-local-subchart.

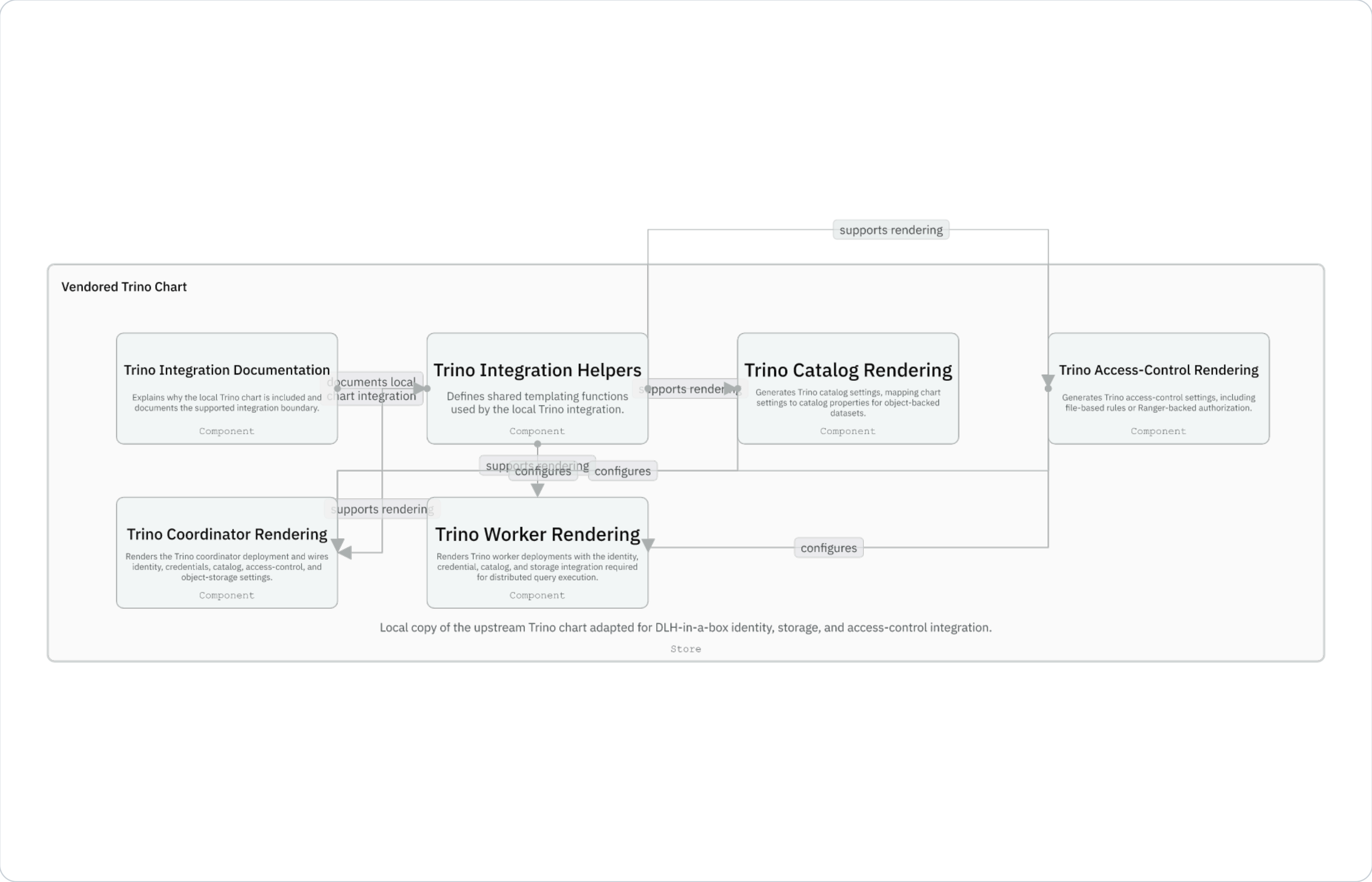
Hive flow from catalog input to metastore runtime



- Catalog iteration is the core pattern.
- Storage wiring supports MinIO or S3.
- PostgreSQL backs metastore state.
- Schema initialization can happen through init containers and hook jobs.

Source: Hive subchart README Mermaid.

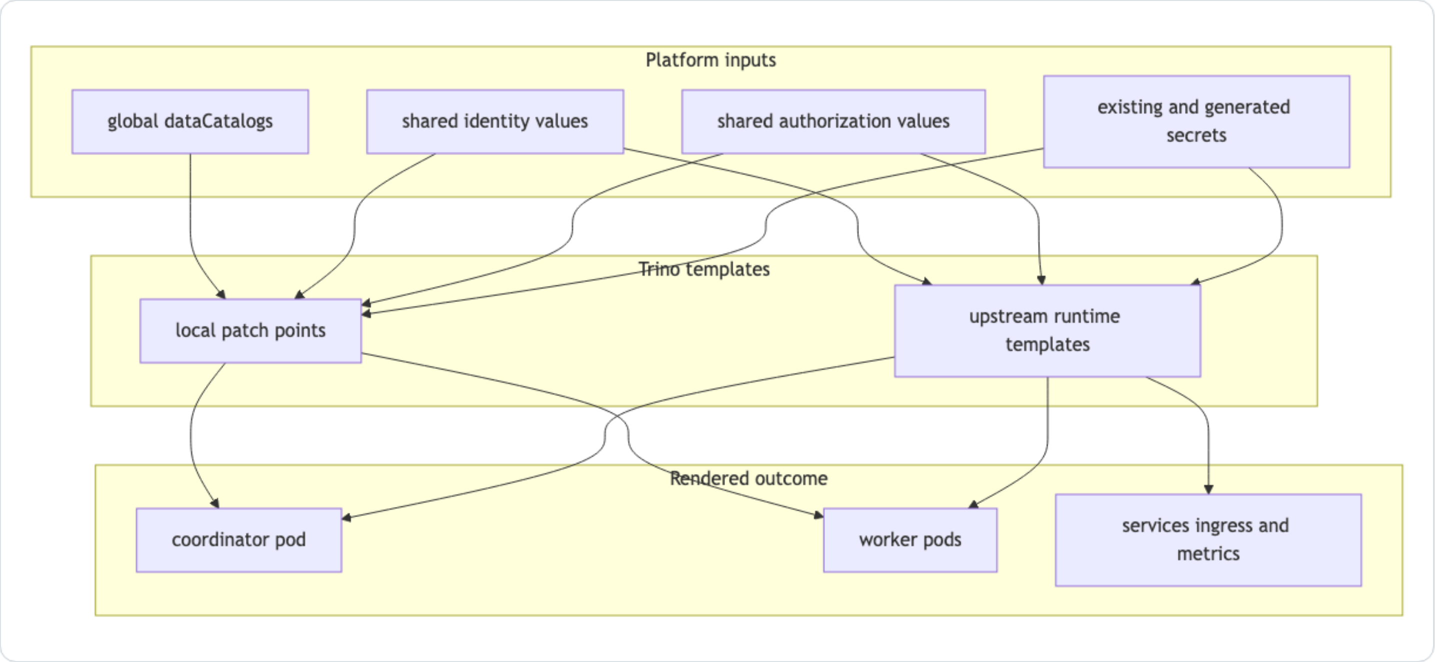
Vendored Trino chart patch points



- Most of the Trino chart is upstream material.
- Local patch points generate catalogs, access-control config, secret mounts, and identity wiring.
- Coordinator and worker deployments consume the generated settings.

Source: IcePanel 06-vendored-trino-chart.

Trino patch model



- Helpers support catalog, access-control, coordinator, and worker rendering.
- Generated catalog config comes from global.dataCatalogs and storage settings.
- Authorization can be file-based or Ranger-backed.

Source: Trino template patch guide Mermaid.

Storage modes

Mode	What changes	Maintainer concern
MinIO	Chart deploys in-cluster S3-compatible storage	Good local default; not a production secret story
External S3	Hive and Trino point at an existing backend	Needs real endpoint, credentials, TLS, and governance context
Existing Secret	Generated catalogs may read credentials via lookup	Offline render and in-cluster upgrade can differ

Source: manual Storage Modes and lookup behavior.

Where DataHub fits

Optional discovery layer

- DataHub is not Trino and not Hive Metastore.
- Its role here is metadata discovery and search.
- The chart carries compatibility templates for auth and prerequisite secret/service shape.
- Deployments decide whether DataHub is part of their runtime.

DataHub is not Trino and not Hive Metastore.

Source: manual Where DataHub Fits; chart guide.

Guided checkpoint: trace a catalog

```
Find global.dataCatalogs in charts/dlh-in-a-
box/values.yaml

Follow it to Hive templates under charts/dlh-in-
a-box/charts/hive/templates

Follow it to Trino catalog rendering under
charts/dlh-in-a-box/charts/trino/templates

Check which example overlays enable MinIO or
external S3
```

Checkpoint questions

- Can the developer name the generated Hive and Trino outputs?
- Can they explain how storage mode changes both paths?

Session 07 landing

What the inheriting team should now be able to do

- They can trace global.dataCatalogs through runtime resources.
- They understand why Hive is first-party and Trino is vendored with patch points.
- They can place DataHub and analysis apps around, not inside, the core query path.

Leave-behind

Use the PPTX notes as a presenter script and source-map.md as the reference trail.